The Physics of the Instant
Analysis from Limits, Relativity, and Renormalization*

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Abstract

Much of theoretical physics rests on the idealization that a physical system possesses a complete state at each instant of time, a state sufficient to determine its future evolution. We argue that this completeness is a physical assumption rather than a consequence of how the instantaneous state is defined, and that established physics provides no basis for it and positive reasons to doubt it. Three independent lines of evidence are assembled. First, operationally the state at a temporal point is the continuous extension of descriptions over finite intervals: the ε-δ definition of a limit assigns a value at the point from its punctured neighborhoods, and whether that value is dynamically sufficient is a property of the dynamics that the construction neither supplies nor guarantees. Second, quantum mechanics and general relativity, applied jointly, bound spatial and temporal resolution at the Planck scale, and the Salecker-Wigner clock bound, combined with the requirement that the clock itself not undergo gravitational collapse, yields the same temporal bound from the properties of the clock rather than of the probed region. These bounds are not theorems of a completed theory of quantum gravity, but within the domain of established physics they show that the limiting process $\delta t \to 0$ cannot be physically implemented. Third, in quantum field theory products of field operators at coincident points are singular, physical predictions are defined at finite resolution, Wilson's analysis identifies this with the scale dependence of effective descriptions rather than with a defect of perturbation theory, and the algebraic formulation assigns observables to extended regions rather than to points. Taken together, these show that completeness at a temporal point is not delivered by the construction of the state, cannot be physically realized, and is not where established theories place their physical content. Descriptions that presuppose it, including Markov dynamics and the Chapman-Kolmogorov composition of transition kernels through intermediate times, inherit the assumption: the memoryless description is the special case that requires justification.

Keywords: instantaneous state; dynamical completeness; temporal resolution; Planck time; Salecker-Wigner clock bound; renormalization group; effective field theory; algebraic quantum field theory; Markov idealization

1 Introduction

The assumption that a physical system possesses a complete state at each instant of time is so deeply embedded in theoretical physics that it is rarely stated as an assumption at all. In

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classical mechanics, the state is specified by positions and momenta at a time t; in quantum mechanics, by a vector in Hilbert space or equivalently a density matrix; in classical field theory, by field configurations and their conjugate momenta on a Cauchy surface. In every case, the formalism presupposes that the state at a temporal point carries the full kinematic specification of the system: all degrees of freedom, all constraints, all information required to determine the system's future evolution.

This paper argues that completeness is not a property the instantaneous state inherits from the way it is constructed, but a separate physical assumption, one for which established physics provides no basis and against which it supplies positive reasons for doubt. Three independent arguments support this. One, from the mathematics of limits, shows that the construction defining the instantaneous state delivers an element of the state space at the point and nothing about its dynamical sufficiency, which is a property of the dynamics that the construction does not address. The second, from relativistic gravitation and quantum clock bounds, shows that the physical process of reaching the temporal point cannot be carried out, so that completeness at the point is an assumption no measurement can underwrite. The third, from quantum field theory, shows that the point-valued limit is exactly where the established formalism becomes singular and that physical content is located at finite resolution. The claim is not that an instant is empty. It is that its completeness must be argued for rather than presupposed, and that the argument runs opposite to the usual direction: the memoryless description, in which the state at each time suffices for all later times, is the case requiring justification.

The assumption has a precise form. The initial-value formulation of dynamics takes data on a single time slice as sufficient to fix the solution; the Markov property takes the state at one time as sufficient to fix the statistics of all later times; the composition of transition kernels through intermediate times, and the notion of an instantaneous collapse in quantum measurement, presuppose the same sufficiency. In each case what is presupposed is not the existence of a state at a temporal point, which is a matter of definition and is not in dispute here, but its dynamical completeness. This paper examines that presupposition directly. Each of the three arguments below draws on established physics; what is new is that they are made to bear jointly on whether completeness at a temporal point is delivered by anything in physics. Consequences for memoryless descriptions, and the connection to the indivisible stochastic processes of Barandes [1, 2, 3], are drawn in Sec. 6 as consequences rather than assumed as motivation.

The question is therefore not whether the state at a temporal point can be *defined*. It can, and nothing here disputes the definition. The question is whether the object so defined inherits the property dynamical frameworks demand of it, the complete kinematic information sufficient to fix future evolution. The construction does not supply that property, and established physics provides independent reasons to doubt it.

2 Argument from Limits

Theories introduce point-indexed states directly: a trajectory $x(t)$, a density operator $\rho(t)$, Cauchy data on a slice. The state of a physical system at a temporal point is nonetheless not directly observable [4]: every physical measurement ever performed has yielded an average over a nonzero interval, and the point-indexed object is confronted with experiment only through such averages. Operationally, the "state at time t" is a mathematical extraction, the limit of interval-averaged states as the interval shrinks to zero. In standard analysis, the infinitesimal interval has no existence; one has either finite intervals or the point, with nothing in between. The question is therefore what the point inherits from the intervals that define it operationally, and in particular whether the completeness attributed to the point-indexed object has any

warrant beyond stipulation.

The ε-δ definition of this limit is mathematically rigorous [5]. The limit value exists and is well defined. This is not in dispute.

What is in dispute is whether the limit point inherits the properties that dynamical frameworks demand of it. Consider what the ε-δ definition actually guarantees. For a function f and a point a, the statement $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all x with $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$. The condition $0 < |x - a|$ is essential: the definition explicitly excludes the point a itself. The limit characterizes the behavior of f in every punctured neighborhood of a while making no assertion about a itself.

This exclusion is not a technicality. It is what makes the limit independent of any value separately assigned at the point, and it is the reason limits are useful: they allow a value to be assigned where the function is not defined, or where its value differs from the limiting behavior. The limit of $\sin(x)/x$ as $x \to 0$ is 1, although the function is undefined at $x = 0$; the continuous extension assigns the value there by stipulation. The construction delivers the value. It delivers nothing else, because nothing else is asked of it.

The physical consequence is direct. The state space may carry whatever structure the theory requires, a phase space or a Hilbert space, and the limit is taken in that space; the limit is then an element of the space with all of the structure that membership confers. What the construction cannot supply is that this element determines the future, because sufficiency is a property of the dynamics, the existence of a law that fixes later states from the present one alone, and not a property of the limit. Two stipulations therefore enter wherever a point-indexed complete state is written down. The first is continuity in time, which identifies the value at the point with the limit of the approach; it is a physical assumption, one that discrete collapse models [33] decline outright and that the algebraic formulation of field theory [29] has no sharp-time object to apply to, since it assigns observables to open regions and none to a sharp time. The second is dynamical sufficiency of the value so identified, which indivisible stochastic dynamics [1] denies. Neither is delivered by the definition of a limit.

That the memoryless description is a property of the limit rather than of any finite-resolution description can be seen directly. Let a process be observed through a temporal window of finite width δ. The observed quantity is a function of the path segment within the window, and a function of a Markov process is generically not Markov [6, 7]: the observed transition family fails to compose through intermediate times for every $\delta > 0$, with a defect that vanishes only as $\delta \to 0$. Divisibility is acquired at the limit and inherited from nothing along the approach. The limit point has a property that no member of the approximating family possesses.

This argument applies within the standard continuum topology of the real line. Alternative topological structures, non-Hausdorff spacetimes in which limits need not be unique [8], or causal set theory in which the continuum is replaced by a discrete partial order [9], would modify the analysis in ways specific to each framework. The present argument targets the standard idealization.

The completeness of the state at a temporal point is therefore not a consequence of how such states are constructed. It is an additional assumption, and the following sections show that established physics gives positive reasons to doubt it.

3 Arguments from Relativity

General relativity and quantum mechanics are among the most precisely confirmed theories in the history of science. Applied jointly, they yield concrete bounds on the resolution of spatial and

temporal measurement through three lines of argument, two for the spatial bound and one, from clock physics, for the temporal one. The bounds are obtained by extrapolating both theories toward a regime where neither is independently known to remain complete; they are robust across the approaches reviewed by Garay [11], but they are not theorems of a completed theory of quantum gravity. We distinguish throughout between a minimum measurable resolution, which is what the arguments establish, and stronger claims of spacetime discreteness or of the failure of point-indexed mathematical states, which they do not.

Mead [10] and Garay [11] establish that a fundamental minimal spatial length must exist as a consequence of the joint requirements of quantum mechanics and general relativity. The argument is forced by the uncertainty relations: through the de Broglie relation, improving spatial resolution to a scale δx requires concentrating momentum of order $\hbar/\delta x$, and therefore energy of order $\hbar c/\delta x$, within the probed region. At sufficiently small δx the Schwarzschild radius of this energy exceeds δx itself, and the region undergoes gravitational collapse. Since the fundamental length deduced by Mead scales as $\sqrt{G}$, the Planck length $l_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-33}$ cm, the unique length constructible from $\hbar$, G, and c, is the natural lower bound, and both analyses yield results consistent with this scale, though neither derives it as a sharp numerical threshold. The existence of this fundamental length in turn limits the precision with which clocks can be synchronized, as Mead shows [10], an observation originally motivated by the problem of infinities in quantum field theory arising from point-particle idealizations, which connects the gravitational argument directly to the renormalization evidence of Sec. 4.

't Hooft's analysis of trans-Planckian scattering [12] arrives at the same spatial obstruction by a route that is arguably more direct: when two particles collide at sufficiently high center-of-mass energy, general relativity requires that the interaction region undergo gravitational collapse. This conclusion depends only on the particle kinematics and the Einstein equations, without recourse to quantum interference effects, and therefore rests on fewer assumptions than the Mead-Garay derivation. The result concerns the scattering configuration; its bearing on temporal resolution is through the inference drawn below, not through any direct statement about pointwise descriptions.

A third route to a temporal bound is provided by Salecker and Wigner [13], who show that quantum mechanics constrains the precision of any clock used to measure spacetime intervals. An ideal clock measuring a time interval T with resolution δt must, to avoid quantum spreading of its center of mass degrading its accuracy, satisfy a minimum mass condition $M \gtrsim \hbar T/(c^2\,\delta t^2)$. This bound alone involves no gravitation. A Planck-scale value arises when it is combined with the requirement that the clock not collapse into a black hole, $r_s = 2GM/c^2 \lesssim \ell_{\rm clock}$, together with the condition $\ell_{\rm clock} \lesssim c\,\delta t$ that the clock be readable within its own resolution. The combination gives

$$\delta t \gtrsim \left(T\,t_P^2\right)^{1/3}, \qquad t_P = \frac{l_P}{c} = \sqrt{\frac{\hbar G}{c^5}} \approx 5.4 \times 10^{-44}\ \text{s}, \tag{1}$$

up to a factor of order unity, the form obtained by Károlyházy [14] and by Ng and van Dam [15], of which $\delta t \gtrsim t_P$ is the corollary for $T \sim \delta t$. The dependence on T is disputed: Diósi and Lukács [16] argue that the bound overstates the uncertainty of spacetime, and Baez and Olson [17] show that a clock attached to a massive body reduces the uncertainty to the Planck scale itself. Only the Planck-scale corollary is used here. The route is independent of the Mead and 't Hooft analyses in that it constrains the clock rather than the probed region, not in that it dispenses with gravitation.

The temporal consequence of the spatial results follows independently. Since the Planck time t_P is the light-crossing time of the Planck length, probing temporal intervals shorter than t_P requires concentrating energy at the same densities that trigger gravitational self-collapse in the Mead and 't Hooft analyses. This temporal extension is not stated by Mead, Garay, or

't Hooft, but is a direct physical inference from their spatial results: the energy scales are identical, related by the factor c. The Mead and 't Hooft analyses, together with this inference, imply that the interaction region collapses into a black hole before the temporal measurement completes. The Salecker-Wigner route arrives at the same temporal bound from a different direction: the quantum-mechanical properties of the clock, combined with the requirement that the clock itself not collapse, prevent arbitrarily fine temporal resolution without any appeal to the collapse of the measured region.

We stress that these bounds follow from the two established theories themselves rather than from the modeling assumptions of any particular approach to quantum gravity, and that they do not rely on the time-energy uncertainty relation, whose interpretation is complicated by the fact that time is not an observable in standard quantum mechanics. The specific physics below t_P is unknown. What is known is that, within the domain of established physics, the limiting process $\delta t \to 0$ cannot be physically implemented. The temporal point is a limit that established physics does not allow to be physically realized.

This is an operational result, and its scope should be stated exactly. It establishes that the point cannot be reached. It does not by itself establish that the mathematical state assigned to the point is incomplete, and no such claim is made here. Its bearing on completeness is this: the direct route by which the completeness of a point-indexed state could be confronted with experiment, the progressive refinement of temporal resolution, is closed at a finite scale. Completeness at the point is therefore an assumption that no measurement can underwrite, and Sec. 2 has shown that it is not underwritten by the construction either.

Approaches to quantum gravity that begin with a discrete structure at the Planck scale, lattice gravity, causal set theory [9], and spin foams, are compatible with this conclusion: they abandon the continuum temporal point from the outset, replacing the standard idealization with a fundamentally different mathematical structure in which the instantaneous state, as conventionally conceived, plays no role.

The preceding section established that the mathematical construction of the point-state does not supply dynamical completeness at the point. This section establishes that the physical process required to reach the point cannot be carried out: gravitational self-collapse and quantum clock bounds independently prevent the limiting process from being physically completed.

4 Argument from Renormalization

Quantum field theory provides corroboration from an independent direction. Quantum fields are operator-valued distributions, and products of field operators at coincident spacetime points are singular. The ultraviolet divergences of perturbation theory are the perturbative expression of this fact: they arise whenever the formalism is pushed to the point-valued limit [18].

The renormalization program [19, 20, 21, 22, 23, 24], given conceptual foundation by Wilson [22, 23, 24], extracts finite physical predictions by introducing a regulator that keeps the theory away from the spacetime point. The physical content is obtained at finite resolution and shown to be independent of the specific regulator. Where a continuum limit exists, it is obtained by removing the regulator from renormalized quantities, not by evaluating the bare theory at a point, and whether an interacting continuum limit exists at all is theory-dependent. For lattice-regularized scalar ϕ^4 theory in four dimensions, the scaling limit at criticality is now known to be Gaussian, that is, trivial [26]: taking the continuum limit removes the interaction. Asymptotically free theories are the case in which an interacting continuum limit is expected. In every case, the predictions that agree with experiment to extraordinary precision [28] are computed from renormalized quantities defined at a finite renormalization scale; where a regu-

lator is removed, it is removed from those quantities, and the bare theory at a point is never evaluated.

It is essential to distinguish the Wilsonian understanding of renormalization from the earlier view in which divergences were regarded as computational artifacts of perturbative methods. Wilson's framework demonstrates that the need for renormalization is not an artifact of perturbation theory. The running of coupling constants with energy scale, for instance the running of the strong coupling α_s in quantum chromodynamics, is an experimentally measured phenomenon [27], not a calculational convenience. In Wilson's picture, a theory at energy scale μ is an effective description whose parameters run with the scale, obtained by integrating out degrees of freedom above that scale. On a lattice of fixed spacing all quantities are finite; renormalization enters in relating descriptions across scales and in taking the continuum limit [25], and the lattice at fixed spacing is itself an effective description whose parameters run. The bare theory at a spacetime point is not an object whose properties are fixed by the effective description; it is the boundary of that description's validity, and its status, where a continuum limit exists at all, is settled theory by theory, as the triviality of ϕ^4 in four dimensions illustrates. The renormalization group is the systematic expression of this fact.

Physics at a spacetime point is where the formalism becomes singular. Physical content is extracted at finite resolution. Whether a future theory could render the spacetime point well defined is an open question; the argument here is evidential, not deductive. Our best established theories, the very theories whose predictions have been confirmed to extraordinary precision, provide no basis for assuming that the point carries complete physical information, and positive grounds for doubting it. This conclusion finds independent support in the algebraic formulation, where the Haag-Kastler axioms assign observable algebras to open spacetime regions [29], and in Wightman's result that fields defined pointwise, rather than as operator-valued distributions, lead to contradictions under natural assumptions [30]. Those results concern the localization of observables; they do not by themselves bear on the composition of the dynamics, which in that framework is carried by the automorphism group. The parallel with the preceding arguments is not analogical but structural: in all three cases, the temporal or spacetime point is where the physics fails, and the meaningful content lives at finite resolution.

5 The Collapse-Timing Formulation of the Measurement Problem

The measurement problem is standardly formulated as the tension between the deterministic, unitary evolution of the quantum state and the occurrence of definite, stochastic outcomes upon measurement [32]. One formulation of this tension presupposes that both sides of the measurement event, the unitarily evolving superposition before and the definite outcome after, are complete states at temporal points, where "complete" means carrying sufficient dynamical information to determine future evolution (whether pure or mixed). The demand is that the timeline can be cut at "the moment of collapse" with a fully specified state on either side. This is precisely the demand that the instantaneous state carry complete dynamical information, applied to the measurement event. We call this the collapse-timing formulation.

Measurement is a physical interaction between system and apparatus. Every stage, photons hitting detectors, electrons cascading through amplifiers, pointers moving on dials, takes nonzero time. There is no instant at which the measurement happens. There is an interval over which system and apparatus become correlated, a picture independently supported by decoherence theory, which models the finite timescale over which system-environment entanglement suppresses interference [31, 32]. At the beginning, they are uncorrelated. At the end, the apparatus registers a definite outcome.

Asking "at which point in between did the transition occur?" is demanding dynamical completeness at intermediate temporal points within the measurement interval. The preceding arguments show this demand is unjustified: the mathematical construction does not supply completeness at the point, the physical limit to the point cannot be reached, and the point-valued limit is where known physics becomes singular.

The collapse-timing formulation of the measurement problem is thereby dissolved rather than solved. The question of the instant at which the transition occurs presupposes complete descriptions at temporal points on either side; without that presupposition the collapse-timing question has no well-defined terms in which to arise. The broader tension between unitary evolution and definite outcomes does not depend on an instant, as the continuous stochastic dynamics of spontaneous localization models make explicit [34, 35], and other aspects of the measurement problem, the derivation of the Born rule, the definite-outcome problem, and the preferred-basis problem, are not addressed by this dissolution and require independent treatment.

The specific outcome of a measurement, why *this* result rather than *that* one, is not addressed by this dissolution, nor does it need to be. Within the framework of indivisible stochastic processes [1, 2, 3], established for systems with finite configuration spaces and unistochastic transition matrices, the outcome is a stochastic draw from the transition matrix over the measurement interval. It is irreducibly stochastic. In the standard Hilbert space formulation, the Schrödinger equation is deterministic, making stochastic outcomes appear to require a mechanism, collapse, that violates the governing equation. In the indivisible stochastic formulation, the dynamics are stochastic from the start; the transition matrix yields probabilities and outcomes are draws from those probabilities, with no deterministic equation being violated. As Barandes establishes [1, 2], the Schrödinger equation deterministically governs the evolution of the transition matrix, the probability law itself, not the realization of individual outcomes, which was never deterministic in either formulation. Within the scope of the correspondence the two formulations are empirically equivalent: they produce identical predictions. The conceptual puzzle is an artifact of the Hilbert space formulation's mathematical structure, not of the physics.

6 Consequences

This is in fact what occurs in practice.

If completeness at a temporal point is an assumption rather than a consequence, then every description that presupposes it is an idealization whose adequacy must be established for the system at hand. Memoryless (Markov) dynamics, and the Chapman-Kolmogorov composition of its transition kernels through intermediate times,

$$\Gamma(t_2 \leftarrow t_1) = \Gamma(t_2 \leftarrow t)\,\Gamma(t \leftarrow t_1), \qquad t_1 < t < t_2, \tag{2}$$

are the principal cases. Two facts fix their status. First, as noted in Sec. 2, for any process observed through a temporal window of finite width the observed transition family is generically non-Markovian [6, 7], and its composition defect vanishes only in the limit of unbounded resolution, a limit that Sec. 3 shows cannot be reached. The memoryless description is therefore a property of the idealized limit and of no finite-resolution description. This is testable: for a system that is Markov on the observed state space, the composition defect of the window-averaged family must fall to zero as the window is refined, whereas a defect that persists under refinement is memory intrinsic to the chosen state space rather than to the resolution. The scaling of the defect with resolution thus separates two sources of Chapman-Kolmogorov failure. Second, Barandes's stochastic-quantum correspondence [1, 2, 3] establishes that quantum systems, described on their configuration spaces, are indivisible stochastic processes whose transition matrices compose through intermediate times only at division events, at any resolution;

the correspondence is established for finite configuration spaces and unistochastic transition matrices, and the claims here are confined to that scope. The burden of justification therefore lies with the memoryless description, whether the system is classical or quantum.

One caution applies. Distinguishing state-space memory of classical origin from quantum indivisibility requires criteria beyond the composition defect and its scaling, since classical hidden variables also produce a resolution-independent defect. Those criteria lie outside the scope of this paper; the quantitative measures of divisibility failure developed for open quantum systems [36, 37] and the process-tensor treatment of classical versus quantum non-Markovianity [38] are the natural starting point.

7 Discussion

The three arguments assembled here draw only on established mathematics and physics, and none invokes a particular interpretation of quantum mechanics, proposes new axioms, or modifies existing theory. The argument from limits requires only the ε-δ definition of a mathematical limit and the elementary fact that a function of a Markov process is generically not Markov. The argument from relativity requires only quantum mechanics and general relativity, each in its established domain and jointly extrapolated to the Planck scale, where gravitational self-collapse and Salecker-Wigner clock bounds independently prevent the temporal point from being physically reached. The argument from renormalization requires only the existence of ultraviolet divergences and their physical interpretation through Wilson's renormalization group. What is new is not any of these results but the use to which they are jointly put: together they remove the basis for treating the completeness of the instantaneous state as given, and they do so without leaving the domain of confirmed physics.

The consequence is a reversal of the burden of proof, and it is a reversal about an assumption, not about the metaphysics of the instant. The completeness of the instantaneous state is not delivered by the construction that defines it, cannot be physically realized, and is not where established theories locate their physical content. It is an assumption, and the descriptions that presuppose it, the initial-value formulation, the Markov property, the composition of transition kernels through intermediate times, and the instantaneous collapse of the quantum state, are idealizations whose warrant must come from elsewhere. Once completeness is no longer presupposed, the memoryless description is the special case that must be justified, and the dissolution of the collapse-timing measurement problem follows as one instance of that reversal rather than as a separate result.

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